MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143E1 ELECTRICAL PARTS AND ASSEMBLIES FOR

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, tests, manufacture and acceptance requirements for Electrical Parts, Fire Board Assemblies and Logic Board Assemblies and Electrical Assemblies (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-27	- Transformer and Inductor, General Specification for
MIL-S-19500	- Semiconductor Devices, General Specification for
MIL-C-39006/22	- Capacitors, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85°C (Voltage Derated to 125°C), Established Reliability, Style CLR79

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-S-48687 - Safety and Arming Device, GM: M143E1 Loading, Assembling and Packing For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-750 - Test Methods for Semiconductor Devices
MIL-STD-1285 - Marking of Electrical and Electronic Parts
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12630829 - Fire Board Assembly
12630845 - Logic Circuit Board Assembly
12630849 - Connector Cable Assembly
12630828 - Electronics Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials, and parts components shall be in accordance with applicable drawings and specifications.

3.2 Assemblies. The complete assemblies shall comply with all requirements specified on Drawings (Dwg.) 12630829, 12630845, 12630828 and 12630849 and associated drawings and with all requirements specified in the applicable specifications and standards.

3.2.1 Measurements (electronic). All measurements required in this specification shall be taken with a measurement device that has as input impedance equal to or greater than 10 megohms and an open circuit potential no greater than 10.0VDC.

3.3 Fire board assembly (Dwg. 12630829).

3.3.1 Continuity. The electrical resistance between the following printed board terminals shall be no greater than 1.0 ohms:

a. E11 to E15
b. E12 to E14
c. E13 to E16

3.3.2 Resistance. The electrical resistance between the ground terminal E1 (except k) to the terminals below shall be as specified in Table I.

TABLE I - Resistance

IDENTIFICATION	TERMINAL	REQUIREMENT (OHMS)
a. Fire Signal Input	E2	46 to 68
b. Detonator Input	E5	1.6 MEG to 2.4 MEG
c. Destruct Input	E6	1.6 MEG to 2.4 MEG
d. Fire TM Input	E7	Greater than 20 MEG
e. Power Input	E8	8.4K to 12.6K
f. ITL Command	E9	Greater than 20 MEG
g. ITL Command Return	E10	Greater than 20 MEG
h. ITL TM	E11	Greater than 20 MEG
i. ITL Monitor	E12	Greater than 20 MEG
j. ITL Monitor	E13	Greater than 20 MEG
k. Resistor, Diode	(+)E3 to (-)E4	2.4K to 19 MEG

3.3.3 Fire circuit characteristics.

3.3.3.1 Fire circuit charging. The fire circuit capacitor voltage shall be 70.0VDC minimum (min.) at (+)E4 to (-)E1 at a maximum (max.) of 2.50 seconds after 21.5 - 0.5VDC is applied to (+)E8 to (-)E1. The initial voltage on (+)E4 to (-)E1 shall be less than 2.5VDC prior to start of test.

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3.3.3.2 Fire circuit capacitor voltage storage. The fire circuit capacitor voltage shall remain 70.0VDC min. at (+)E4 to (-)E1 six (6) seconds min. after the 21.5 - 0.5VDC charging voltage is removed from (+)E8 to (-)E1.

3.3.3.3 No charging. With a 100 ± 10%, 1/4 watt min. resistor connected between E5 and E1, the fire circuit capacitor voltage (+)E4 to (-)E1 shall be no greater than 15VDC after applying an 18 ± 2.0V pulse with a duration of 500 ± 100msec at (+)E2 to (-)E1. Within 100 + 10msec of the application of the 18 ± 2.0V pulse apply 28 ± 2.8VDC at (+)E8 to (-)E1 for a period not to exceed 10 - 5 sec.

3.3.3.4 Fire circuit stabilized voltage. The fire circuit capacitor voltage shall be between 70.0VDC min. to 105.0VDC max. at (+)E4 to (-)E1 after a min. of 10 seconds charging with 21.5 - 0.5VDC applied at (+)E8 to (-)E1.

3.3.4 Fire circuit function. With a 100 ohms ± 10%, 1/4 watt min. resistor connected between E5 and E1, the fire circuit is defined as having functioned when the fire circuit capacitor voltage (+)E4 to (-)E1 has discharged from 70.0VDC min. to less than 15.0VDC in no greater than 400msec after a fuze fire signal or logic destruct signal is applied.

3.3.4.1 Logic destruct. The fire circuit shall function when 1.0 ± 0.2VDC is applied at (+)E6 to (-)E1 for 100 ± 20msec.

3.3.4.2 No fire signal A. The fire circuit shall not function when a 5.0 ± 0.25V pulse with a duration of 200 ± 10 microseconds with a rise time no greater than one (1) microsecond is applied at (+)E2 to (-)E1.

3.3.4.3 No fire signal B. The fire circuit shall not function when a 10.0 ± 0.5V pulse with a duration of 2.0 + 0.5 microseconds with a rise time no greater than one (1) microsecond is applied at (+)E2 to (-)E1.

3.3.4.4 No fire signal C. The fire circuit shall not function when a 30.0 ± 1.5V pulse with a duration of 0.4 ± 0.05 microseconds with a rise time no greater than 0.08 microseconds is applied at (+)E2 to (-)E1.

3.3.4.5 Fuze fire signal. The fire circuit shall function when a fuze fire signal pulse of 18.0 - 2.0V with a duration of 90 - 10 microseconds with a rise time no greater than 20 microseconds is applied at (+)E2 to (-)E1.

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3.4 Logic circuit board assembly (Dwg. 12630845). Prior to start and for duration of testing, the following load and biasing resistors shall be added externally to the logic circuit board under test.

PIN (+) TO (-)	RESISTOR VALUE (OHMS)
E26 - E3	100K
E25 - E3	100K
E23 - E3	1K
E22 - E3	1K (2 watt min.)
E1 - E3	47K
E2 - E3	47K
E6 - E3	47K
E24 - E3	1K

Unless otherwise specified, all resistor are 1/4 watt min., ± 10%.

3.4.1 Continuity and resistance. The electrical resistance shall be as specified in Table II.

TABLE II - Continuity and resistance

TERMINAL (+)	TERMINAL (-)	REQUIREMENTS (OHMS)	TERMINAL (+)	TERMINAL (-)	REQUIREMENTS (OHMS)
E4	E3	1.0 max.	E13	E12	120K to 240K
E29	E3	90 to 110	E19	E3	0.9 to 1.1 MEG
E30	E3	90 to 110	E18	E3	0.9 to 1.1 MEG
E31	E15	90 to 110	E15	E3	0.9 to 1.1 MEG
E20	E21	Less than 19 MEG	E12	E3	88K to 132K
			E13	E3	88K to 132K
E28	E3	80K to 120K	E16	E32	0.99 to 1.01 MEG
E27	E12	120K to 240K	E17	E33	0.99 to 1.01 MEG
E27	E13	120K to 240K	E21	E20	Greater than 20 MEG
E12	E13	120K to 240K	E5	E3	Greater than 20 MEG

3.4.2 Charge command test.

3.4.2.1 TM monitors. With 100 + 5.0VDC applied at right Convertor input (+)E25 to (-)E3 and followed by 31.0 - 0.5VDC applied at Charge Command input (+)E2 to (-)E3, after 0.5sec min. the voltage between the indicated terminals to (-)E3 shall be as specified in Table III.

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TABLE III - TM monitors

	IDENTIFICATION	TERMINAL (+)	REQUIREMENT (volts)
a.	Power/Destruct	E1	-0.5 to +0.5
b.	5VDC Test	E6	-0.5 to +0.5
c.	ITL TM	E8	3.5 to 5.5
d.	Fire TM	E9	-0.5 to +0.5
e.	Arm TM	E10	3.5 to 5.5
f.	Rotor Monitor, Right	E12	3.5 to 5.5
g.	Rotor Monitor, Left	E13	3.5 to 5.5
h.	5VDC (Regulator) Monitor	E14	3.5 to 5.5
i.	Power Output (Diode Drop)	E22	28.5 to 31.0
j.	Convertor Input, Left	E26	-0.5 to +0.5
k.	ITL Unlock	E27	3.5 to 5.5
l.	5VDC	C1(+)	3.5 to 5.5
m.	5VDC	C2(+)	3.5 to 5.5

3.4.2.2 Fire TM. With the voltages applied as specified in 3.4.2.1, the Fire TM voltage at (+)E9 to (-)E3 shall be 4.5 ± 1.0VDC when 5.0 ± 0.5VDC is applied at (+)E28 to (-)E3.

3.4.2.3 Arm TM. With the voltages applied as specified in 3.4.2.1, the Arm TM voltage at (+)E10 to (-)E3 shall be 0 ± 0.5VDC when E12 is connected to E3. With E12 removed from E3 and E13 connected to E3, the Arm TM shall be 0 ± 0.5VDC.

3.4.2.4 ITL TM. With the voltages applied as specified in 3.4.2.1, the ITL TM voltage at (+)E8 to (-)E3 shall be 0 ± 0.5VDC when a 100 ohm ± 10 percent resistor is connected between E27 and E3 ground. Remove the 100 ohm resistor at completion of measurement.

3.4.2.5 Convertor TM. Apply the Charge Command voltage and remove the right convertor voltage as specified in 3.4.2.1. After 0.5sec min. the voltage at (+)E11 to (-)E3 shall be 0 ± 0.5VDC and the Destruct Voltage (+)E23 right and (+)E24 left to (-)E3 shall be 1.5VDC min.

3.4.2.6 Destruct pulse.

3.4.2.6.1 Power up. The voltage on both 68 microfarad capacitors shall be less than 0.5VDC prior to the start of this test. After the voltage is applied as specified in 3.4.2.1, the destruct pulse at (+)E23 right and (+)E24 left to (-)E3 shall change from 0 ± 0.5VDC to 1.5VDC min. in less than 50msec. The duration of the Destruct pulse shall be not less than 2 msec and not greater than 200msec.

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3.4.2.6.2 Convertor input removed. With the voltage applied as specified in 3.4.2.1, for 0.5sec min., the Destruct pulse at (+)E23 right and (+)E24 left to (-)E3 shall be 1.5VDC min. in not less than 10msec and not greater than 50msec after the right convertor voltage is removed.

3.4.2.7 Capacitor storage. Apply 5.0 + 1.0VDC at 5VDC Test (+)E6 to (-)E3, 100 + 5.0VDC at converter input (+)E25 to (-)E3, and 31 - 0.5VDC at Charge Command (+) E2 to (-)E3 and Logic Board Power output (+)E22 to (-)E3. These voltages shall be applied for not less than two (2) seconds prior to this test. Remove the voltage at (+)E2 to (-)E3 and after a 10 ± 1 second interval remove the voltages at (+)E6 to (-)E3 and (+) E22 to (-)E3. The output (Destruct Pulse) at (+)E23, Right and (+)E24, Left to (-)E3 shall be 1.0VDC min. for 50 milliseconds min. upon removal of voltage at (+)E6 to (-)E3.

3.4.3 Power/destruct tests.

3.4.3.1 TM monitors. With 100 + 5.0VDC applied to Left Convertor (+)E26 to (-)E3 and 23.0 + 0.5VDC applied to Power/ Destruct (+)E1 to (-) E3, the voltage across the terminals to (-)E3 shall be as specified in Table IV.

TABLE IV - TM monitors

	IDENTIFICATION	TERMINAL (+)	REQUIREMENT (volts)
a.	Charge Command	E2	-0.5 to +0.5
b.	5VDC Test	E6	-0.5 to +0.5
c.	ITL TM	E8	3.5 to 5.5
d.	Fire TM	E9	-0.5 to +0.5
e.	Arm TM	E10	3.5 to 5.5
f.	Rotor Monitor, Right	E12	3.5 to 5.5
g.	Rotor Monitor, Left	E13	3.5 to 5.5
h.	5VDC (Regulator) Monitor	E14	3.5 to 5.5
i.	Power Output (Diode Drop)	E22	21.0 to 22.5
j.	Converter Input, Right	E25	-0.5 to +0.5
k.	ITL TM	E27	3.5 to 5.5
l.	5VDC	C1(+)	3.5 to 5.5
m.	5VDC	C2(+)	3.5 to 5.5

3.4.3.2 Power TM threshold. With 28 ± 3VDC initially applied to Power/Destruct (+)E1 to (-)E3 and the left convertor voltage applied as specified in 3.4.3.1, the Power TM voltage at (+)E7 to (-)E3 shall be 4.5 ±1.0VDC, when the Power/Destruct input applied at (+)E1 to (-)E3 is equal to or greater than 25.0VDC. The Power TM voltage at (+)E7 to (-)E3 shall be 0 ± 0.5VDC, when the Power/ Destruct input applied to (+)E1 to (-)E3 is less than 21.0VDC. This transition shall occur at 23.0 ± 2.0VDC.

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3.4.3.3 Destruct pulse threshold. With the voltages initially applied as specified in 3.4.3.1, the Destruct voltage at both terminals (+)E23 right and (+)E24 left to (-)E3 shall be 0 ± 0.5VDC when the Power/Destruct input applied at (+)E1 to (-)E3 is equal to or greater than 20.5VDC. The Destruct voltage at both (+)E23 right and (+)E24 left to (-)E3 shall be 1.5VDC min. when the Power/Destruct input applied at (+)E1 to (-)E3 is less than 18.5VDC. This transition shall occur at 19.5 ± 1.0VDC.

3.4.3.4 Converter threshold. With the voltages initially applied as specified in 3.4.3.1, the Destruct voltage at both (+)E23 right and (+)E24 left to (-)E3 shall be 0 ± 0.5VDC and the convertor TM at (+)E11 to (-)E3 shall be 4.5 ± 1.0VDC when the left convertor voltage (+)E26 to (-)E3 is greater than 70.0VDC. The destruct voltage at (+)E23 right and (+)E24 left to (-)E3 shall be 1.5VDC min. and the Convertor TM at (+)E11 to (-)E3 shall be 0 ± 0.5VDC when the voltage on (+)E26 to (-)E3 is less than 60.0VDC. This transition shall occur at 65.0 ± 5VDC.

3.4.4 Five-volt test. Apply 100 + 5.0VDC to left convertor (+)E26 to (-)E3, 23.0 + 0.5VDC to Power/Destruct (+)E1 to (-)E3, 23.0 + 0.5VDC to logic board power output (+)E22 to (-)E3 and 5.0 ± 0.5VDC to 5VDC test (+)E6 to (-)E3 for the following test.

3.4.4.1 Destruct pulse. Wait a min. of 0.5 seconds then remove the 23VDC at E1 followed by removing the 5VDC at E6 as specified in 3.4.4. The Destruct voltage at (+)E23 right and (+)E24 left to (-)E3 shall change from 0 ± 0.5VDC to 1.5VDC min. in not less than 10msec and not greater than 50msec.

3.5 Connector cable assembly - (Dwg 12630849).

3.5.1 Continuity and isolation. When each of the points listed below is activated with a logic level high, the corresponding points indicated shall be at a logic level high and all others shall be at a logic level low. Abbreviations are as follows: J1 = connector pin; L = logic board connection; FR = right fire board connection; FL = left fire board connection; SR = right rotor switch connection; SL = left rotor switch connection.

Parts Activated	Points Requiring Logic High
J1-1	L6
J1-2	L13, SL1
J1-3	FR13, L18
J1-4	L8
J1-5	FL13, L19
J1-6	J1-34

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Parts Activated	Points Requiring Logic High
J1-7	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-8	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2,
J1-9	All Low
J1-10	All Low
J1-11	All Low
J1-12	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2,
J1-13	J1-45, J1-40, J1-39, J1-29, J1-24, J17, J1-16, J1-15, J1-14, J1-12, J1-8, J1-7,L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2,
J1-14	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-15	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-16	J1-45, J1-40, J1-39, J1-29, J1-24, J1-17, J1-115, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-17	J1-45, J1-40, J1-39, J1-29, J1-24, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7,L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-18	L12, SR1
J1-19	L10
J1-20	L9
J1-21	L7
J1-22	All Low
J1-23	L14

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Parts Activated	Points Requiring Logic High
J1-24	J1-45, J1-40, J1-39, J1-29, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-25	All Low
J1-26	J1-42, FR2
J1-27	J1-43, FR12
J1-28	All Low
J1-29	J1-45, J1-40, J1-39, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-30	J1-46, L1
J1-31	J1-47, L2
J1-32	J1-48, L21, FR9, FL9
J1-33	J1-49, FR10, FL10
J1-34	J1-6
J1-35	L15
J1-36	L16
J1-37	L11
J1-38	L17
J1-39	J1-45, J1-40, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-40	J1-45, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-41	FR1A
J1-42	J1-26, FR2
J1-43	J1-27, FL2
J1-44	FL1A
J1-45	J1-40, J1-39, J1-29, J1-24, J1-17, J1-16, J1-15, J1-14, J1-13, J1-12, J1-8, J1-7, L4, L3, FR1, FL1, SR6, SR3, SR2, SL6, SL2
J1-46	J1-30, L1
J1-47	J1-31, L2
J1-48	J1-32, L21, FR9, FL9
J1-49	J1-33, L20, FR110, FL10
J1-50	L5, Chassis Ground (Hole #182)
Case Ground (Hole #182)	All Low
L22	FR8, FL8
L23	FR6

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Parts Activated	Points Requiring Logic High
L24	FL6
L25	FR3
L26	FL3
L27	FR11, FL11
L28	FR7, FL7
L29	FR12
L30	FL12
L31	SL3
L32	FR4
L33	FL4
SR4	SL4
SR5	FR5
SL5	FL5

3.6 Electrical assembly (EA) (Dwg. 12630828). S&A mechanical switching shall be simulated through the use of electrical connections external to the electrical assembly as shown in Table V.

TABLE V - External connections

IDENTIFICATION	SAFE & LOCKED	SAFE & UNLOCKED	ARMED & LOCKED
S1(L&R)E1-E2	Open	Closed	Open
S1(L&R)E1-E3	Closed	Open	Closed
S2(L&R)E1-E2	Open	Open	Closed
S2(L&R)E3-E4	Closed	Closed	Open
S2(L&R)E5-E6	Open	Open	100 ohms*

*A 100 ohms = ± 10%, 1/2 watt min. resistor shall be connected between the points indicated.

3.6.1 Continuity. The electrical resistance between the pins specified in Table VI with the EA in any position shall be no greater than 1.0 ohm while applying an open electrical potential no greater than 10.0VDC.

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TABLE VI - Continuity

J1-14 to J1-7	J1-14 to J1-41
J1-14 to J1-8	J1-14 to J1-44
J1-14 to J1-12	J1-14 to J1-45
J1-14 to J1-13	J1- 6 to J1-34
J1-14 to J1-15	J1-26 to J1-42
J1-14 to J1-16	J1-27 to J1-43
J1-14 to J1-17	J1-30 to J1-46
J1-14 to J1-24	J1-31 to J1-47
J1-14 to J1-29	J1-32 to J1-48
J1-14 to J1-39	J1-33 to J1-49
J1-14 to J1-40	J1-50 to Frame Ground Pad (Hole #182)

3.6.2 Resistance. The electrical resistance shall be as specified in Table VII with the EA in the position indicated while applying an open electrical potential no greater than 10VDC.

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TABLE VII - Resistance

IDENTIFICATION	J-1 CONNECTOR PINS (+) to (-)	POSITION AND REQUIREMENT (OHMS) SAFE & LOCKED	SAFE & UNLOCKED	ARMED & LOCKED
a. Condition Monitor	35 to 15	90 to 110	90 to 110	.90M to 1.1M
b. Latch Monitor, Right	3 to 15	90 to 110	.90M to 1.1M	90 to 110
c. Latch Monitor, Left	5 to 15	90 to 110	.90M to 1.1M	90 to 110
d. Fuze Fire Signal, Right	26 to 15	46 to 68	46 to 68	46 to 68
e. Fuze Fire Signal, Left	27 to 15	46 to 68	46 to 68	46 to 68
f. Rotor Monitor, Left	2 to 15	88K to 132K	88K to 132K	*NGT 1.0
g. Rotor Monitor, Right	18 to 15	88K to 132K	88K to 132K	*NGT 1.0
h. Power/Destruct to ITL Command	30 to 32	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
i. Power/Destruct to Charge Command	30 to 31	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
j. Charge Command to ITL Command	31 to 32	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
k. Charge Command to Power/Destruct	31 to 30	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm
l. Ground to ITL Return	14 to 33	Greater than 20 megohm	Greater than 20 megohm	Greater than 20 megohm

*NGT - Not greater than

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3.6.3 Connector pins to case ground isolation resistance. The electrical resistance between J1-50 to the pins specified in Table VIII with the EA in any position shall be greater than 20 megohms while applying an open electrical potential no greater than 10.0VDC.

TABLE VIII - Isolation resistance

J1-1	J1-18	J1-34
J1-2	J1-19	J1-35
J1-3	J1-20	J1-36
J1-4	J1-21	J1-37
J1-5	J1-22	J1-38
J1-6	J1-23	J1-39
J1-7	J1-24	J1-40
J1-8	J1-25	J1-41
J1-9	J1-26	J1-42
J1-10	J1-27	J1-43
J1-11	J1-28	J1-44
J1-12	J1-29	J1-45
J1-13	J1-30	J1-46
J1-14	J1-31	J1-47
J1-15	J1-32	J1-48
J1-16	J1-33	J1-49
J1-17		

3.6.4 Five volt test. With 5.0 + 1.0VDC applied to 5V DC test input (+)J1-1 to (-)J1-15, the voltage across the pins specified in Table IX shall be as follows with the EA in the position indicated.

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TABLE IX - Five-volt test

IDENTIFICATION	CONNECTOR PINS To (-) J1-15	SAFE & LOCKED	SAFE & UNLOCKED	ARMED & LOCKED
a. Rotor Monitor, Left	J1-2	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
b. ITL TM	J1-4	4.5 ± 1.0VDC	0 ± 0.5VDC	4.5 ± 1.0VDC
c. Rotor Monitor, Right	J1-18	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
d. ARM TM	J1-19	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
e. FIRE TM	J1-20	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
f. Converter TM	J1-37	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
g. Capacitor Monitor, Left	J1-38	2.5VDC MAX	2.5VDC MAX	2.5VDC MAX
h. Capacitor Monitor, Right	J1-36	2.5VDC MAX	2.5VDC MAX	2.5VDC MAX

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3.6.5 Operating input voltage. The EA shall meet all electrical output requirements herein specified for all input voltages greater than or equal to 22.5VDC and less than or equal to 31.5VDC on the Power Destruct (+)J1-30 to (-)J1-15 or the Charge Command (+)J1-31 to (-)J1-15 or both.

3.7 Fire circuit characteristics.

3.7.1 Fire circuit charging (charge command). With the EA in the Safe and Unlocked position, the fire circuit capacitor voltage shall be 70.0VDC minimum (min.) at (+)J1-36 (Right) and (+)J1-38 (Left) to (-)J1-15 at a maximum (max.) of 2.50 seconds after 25.2 - 0.5VDC is applied to Charge Command (+)J1-31 to (-)J1-15. The initial voltage on (+)J1-36 and (+)J1-38 to (-)J1-15 shall be less than 2.5VDC prior to start of test.

3.7.1.1 Fire circuit capacitor voltage storage. With the EA in the Safe and Unlocked position, the fire circuit capacitor voltage shall remain 70.0VDC minimum (min.) at (+)J1-36 (Right) and (+)J1-38 (Left) to (-)J1-15 for six (6) seconds min., after the 25.2 - 0.5VDC charging voltage is removed from (+)J1-31 to (-)J1-15.

3.7.2 Fire circuit charging (power/destruct). With the EA in the Safe and Unlocked position, the fire circuit capacitor voltage shall be 70.0VDC minimum (min.) at (+)J1-36 (Right) and (+)J1-38 (Left) to (-)J1-15 at a maximum of 2.50 seconds after 25.2 - 0.5VDC is applied to (+)J1-30 to (-)J1-15. The initial voltage on (+)J1-36 and (+)J1-38 to (-)J1-15 shall be less than 2.5VDC prior to start of test.

3.7.3 No charging while armed. With the EA in the Armed and Locked position, the voltage on the firing circuit capacitors shall be no greater than 15.0VDC at (+)J1-36 (Right) and (+)J1-38 (Left) to (-)J1-15 after 28.0 $\pm$ 2.8VDC is applied for 6 $\pm$ 4 sec to Charge Command (+)J1-31 to (-)J1-15.

3.7.4 Fire circuit function. The left or right fire circuit is defined as having functioned at the time the Fire TM output changes from 0 $\pm$ .5VDC to 4.5 $\pm$ 1.0VDC at (+)J1-20 to (-)J1-15. The left fire circuit has functioned when the left fire circuit capacitor voltage (+)J1-38 to (-)J1-15 has discharged to less than 15.0VDC in no greater than 400msec after the Fire TM output pulse first changes from 0 $\pm$ .5VDC to 4.5 $\pm$ 1.0VDC. The right fire cuircuit has functioned when the right fire circuit capacitor voltage (+)J1-36 to (-)J1-15 has discharged to less than 15.0VDC in no greater than 400msec after the Fire TM output pulse has been detected.

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3.7.4.1 Logic destruct.

3.7.4.1.1 Loss of power after arming. The fire circuits shall function in no less than 10ms and no greater than 50ms.

3.7.4.1.2 Fire upon arming after loss of power. The fire circuits shall function upon arming after loss of power.

3.7.4.1.3 Power destruct threshold. The fire circuit shall function when the voltage applied to Power/Destruct (+)J1-30 to (-)J1-15 is between 18.0 and 21.0VDC.

3.7.4.2 Fuze fire function.

3.7.4.2.1 No fire signal A. Neither fire circuit shall function when a 5.0 ± 0.25V pulse for a duration of 200 ± 10 microseconds with a rise time no greater than one (1) microsecond is applied to (+)J1-27 (Left) and (+)J1-26 (Right) to J1-15 fuze signal input. At completion of the No Fire signal both fire circuit capacitor monitors shall be 70.0VDC min.

3.7.4.2.2 No fire signal B. Neither fire circuit shall function when a 10.0 ± 0.5V pulse for a duration of 2.0 + 0.5 microseconds with a rise time no greater than one (1) microsecond is applied to (+)J1-27 (Left) and (+)J1-26 (Right) to (-)J1-15 fuze signal input. At completion of the No Fire signal both fire circuit capacitor monitors shall be 70.0VDC min.

3.7.4.2.3 No fire signal C. Neither fire circuit shall function when a 30.0 ± 1.5V pulse for a duration of 0.4 ± 0.05 microseconds with a rise time no greater than 0.08 microseconds is applied to (+)J1-27 (Left) and (+)J1-26 (Right) to (-)J1-15 fuze signal input. At completion of the No Fire signal both fire circuit capacitor monitors shall be 70.0VDC min.

3.7.4.2.4 Fuze fire signal. Both fire circuits shall function when a fuze fire signal of 18.0 - 2.0V for a duration of 90 - 10 microseconds with a rise time NGT 20 microseconds is applied to (+)J1-27 (Left) and (+)J1-26 (Right) to (-)J1-15 .

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3.8 EA interface outputs (J-1 connector).

3.8.1 Power TM threshold. The Power TM monitor at (+)J1-21 to (-)J1-15 shall be 4.5 ± 1.0VDC when the Power/Destruct input at (+)J1-30 to (-)J1-15 is greater than 25.0VDC. The Power TM monitor shall be 0 ± 0.5 when the Power/Destruct input is less than 21.0VDC. This transition shall occur at 23.0 ± 2.0VDC.

3.8.2 Voltage outputs. With the EA in the position indicated in Table X the voltage requirements shall be as specified when the input voltage as indicated is applied to Power/Destruct (+)J1-30 to (-)J1-15 for ten (10) seconds min. prior to measurements.

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TABLE X - Interface outputs (J1)

IDENTIFICATION	CONNECTOR PINS To (-) J1-15	INPUT VOLTAGE POWER/DESTRUCT (23.0 - 0.5VDC) SAFE & LOCKED	(23.0 - 0.5VDC) SAFE & UNLOCKED	(31.0 + 0.5VDC) ARMED & LOCKED
a. Rotor Monitor, Left	J1-2	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
b. ITL TM	J1-4	4.5 ± 1.0VDC	0 ± 0.5VDC	4.5 ± 1.0VDC
c. Rotor Monitor, Right	J1-18	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
d. ARM TM	J1-19	4.5 ± 1.0VDC	4.5 ± 1.0VDC	0 ± 0.5VDC
e. FIRE TM	J1-20	0 ± 0.5VDC	0 ± 0.5VDC	0 ± 0.5VDC
f. 5VDC Monitor	J1-23	4.5 ± 1.0VDC	4.5 ± 1.0VDC	4.5 ± 1.0VDC
g. 100V Capacitor Monitor (Right)	J1-36	70 to 110VDC	70 to 110VDC	80 to 110VDC
h. Converter TM	J1-37	4.5 ± 1.0VDC	4.5 ± 1.0VDC	4.5 ± 1.0VDC
i. 100V Capacitor Monitor (Left)	J1-38	70 to 110VDC	70 to 110VDC	80 to 110VDC

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3.9 Environmental and mechanical.

3.9.1 Temperature cycling. Semiconductors, when tested as specified in MIL-STD-750, Method 1051.1 and MIL-STD-202, Method 107, Test Condition F-1, shall meet the final electrical requirements after exposure.

3.9.2 Constant acceleration. Semiconductors, when tested in the Y1 and Z1 axes in accordance with MIL-STD-750, Method 2006, at 20000g's, shall meet the final electrical requirements.

3.9.3 Seal.

3.9.3.1 Fluorocarbon gross leak. Semiconductors shall meet the requirements specified in MIL-STD-750, Method 1071, Test Condition C.

3.9.3.2 Fine leak. Semiconductors shall meet the requirements specified in MIL-STD-750, Method 1071, Test Condition H or G.

3.9.4 Burn-in. Semiconductors shall be subjected to burn-in testing in accordance with the details herein and MIL-STD 750, Method 1038 or 1039, Test Condition B, for 168 hours at 125°C minimum, and shall meet the final electrical requirements after exposure.

3.9.5 Thermal shock. Semiconductors, when tested as specified in MIL-STD-750, Method 1056, Test Condition B, 15 cycles, shall meet the final electrical requirements after exposure.

3.9.6 Mechanical shock. Semiconductors, when subjected to the shock tests specified in MIL-STD-750, Method 2016, 1500g's, 0.5 milliseconds, 6 shock pulses X1, Y1, and Z1 axes, shall meet the final electrical requirements after exposure.

3.9.7 Life cycle. Transformers and inductors, when subjected to the life cycle tests specified in MIL-T-27, 168 hours at 125°C shall meet the final electrical requirements after exposure.

3.10 Marking.

3.10.1 Lot identification code. Components specified in Table XI shall be marked by a code indicating the week inspection including burn-in was completed for the lot accumulation period. The marking shall be as specified in MIL-STD-1285.

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TABLE XI - Lot identification code

DRAWING	Part Description
12630829	Capacitor, 86 microfarad
12630831	Choke
12630840	Transformer
12630829	SCR, 2N 2326 JANTX
12630860	Bridge Rectifier
12630838	Transistor, 2N6193
12630845	Capacitor, 68 Microfarad

3.10.2 Test data. Test data from tests specified in 4.4.2.4.1 and 4.4.2.15.1 shall be submitted if requested by the procuring activity. Variable data shall be submitted for those requirements specified in 3.3.2, 3.3.3.1, 3.3.3.2, 3.3.3.4, 3.3.4, 3.4.1, 3.4.2.1 thru 3.4.2.5, 3.4.2.6.1, 3.4.2.6.2, 3.4.2.7, 3.4.3.1, 3.4.3.2, 3.4.3.3, 3.4.3.4, 3.4.4.1, 3.6.2, 3.6.4, 3.7.1, 3.7.1.1, 3.7.2, 3.7.3, 3.7.4, 3.7.4.1.1, 3.7.4.1.2, 3.7.4.1.3, 3.8.1 and 3.8.2. Attribute data (go, no-go) may be submitted for all other requirements.

3.11 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.12 Workmanship. All parts and assemblies shall be fabricated in a careful and workmanlike manner. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with

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all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Drawings	Part Description	Quantity
12630828	Electronics Assembly	5
12630829	Capacitor, 86 microfarad	5
12630829	SCR, JANTX 2N2326	5
12630829	Semiconductor, Silicon Diode JANTX IN485B	5
12630829	Fire Board Assembly	5
12630839	Fire Circuit Board	5
12630831	Choke	5
12630840	Transformer	5
12630860	Bridge Rectifier	5
12630838	Transistor 2N6193	5
12630857	Connector	5
12630851	Flexible Cable	5
12630846	Logic Circuit Board	5
12630845	Logic Circuit Board Assembly	5
12630845	Capacitor, 68 microfarad	5
12630849	Connector Cable Assy, prior to potting	5
12630850	Molding Compound	5
12630849	Connector Cable Assembly	5
12630705	Mounting Plate (rear)	5
12630855	Plate, Mounting Connector	5
12630854	Connector Assy	5

4.3.1.1 Homogeneity. The first article shall comply with the homogeneity criteria of MIL-A-48078, 4.4.1.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, componment, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

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TABLE XII - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See 4.3.1
4.3.4	Electrical Parts and Assemblies; M143E1				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Assemblies, Subassemblies and Components (Drawing list - See 4.3.1) Examination for Defects	See 4.3.1		3	4.4.2.1 thru 4.4.2.27

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Tests. Paragraphs, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Level II.

4.4.3 Testing. N/A

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630829
4.4.2.1	Capacitor (86 Microfarad)				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	Capacitance Bridge/ MIL-C-39006/22-0200
102	Dissipation factor		0.40%	3.2	Capacitance Bridge/ MIL-C-39006/22-0200
Minor					
201	Marking missing or incorrect		0.65%	3.7	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Semiconductor, Device Silicon Diode JANTX IN485B				12630829/MIL-S-19500
					NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Band missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630829/MIL-S-19500
4.4.2.3	Rectifier, Controlled, Silicon JANTX2N2326				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Marking		0.65%	3.7	Visual
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630829
4.4.2.4	Fire Board Assembly (Visual)				NEXT HIGHER ASSEMBLY 12630828
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering not in accordance with MIL-STD-1460		0.40%	3.2	Visual
102	Capacitor or diode polarity incorrect		0.40%	3.2	Visual
103	Incorrect component, damaged or missing component		0.40%	3.2	Visual
104	Maximum height of component leads and soldering		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630829
4.4.2.4.1	Fire Board Assembly (Electrical Test)				NEXT HIGHER ASSEMBLY 12630828
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Continuity		100%	3.3.1	Test
102	Resistance		100%	3.3.2	Test
103	No charging		100%	3.3.3.3	Test
104	Fire circuit charging		100%	3.3.3.1	Test
105	Fire circuit capacitor storage		100%	3.3.3.2	Test
106	Fuze Fire Signal		100%	3.3.4.5	Test
107	No Fire Signal "A"		100%	3.3.4.2	Test
108	No Fire Signal "B"		100%	3.3.4.3	Test
109	No Fire Signal "C"		100%	3.3.4.4	Test
110	Logic destruct signal		100%	3.3.4.1	Test
111	Fire circuit stabilized voltage		100%	3.3.3.4	Test

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630839
4.4.2.5	Fire Circuit Board				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole locations		0.40%	3.2	Gage
102	Hole diameters		0.40%	3.2	Gage
103	Board cracked, chipped, warped or twisted		0.40%	3.2	Visual
104	Conductive paths reduced by 1/3 or more of the width		0.40%	3.2	Visual
105	Conductor not bonded to board or peeling		0.40%	3.2	Visual
106	Bridging of conductors		0.40%	3.2	Visual
Minor					
201	Conductor areas not plated		0.65%	3.2	Visual
202	Thickness		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630831/MIL-T-27
4.4.2.7	Choke (Visual)				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering not in accordance with MIL-STD-1460 (2 places) (prior to potting))		0.40%	3.2	Visual
102	Cup height, max. (after potting)		0.40%	3.2	Gage
Minor					
201	Marking (after final electrical)		0.65%	3.7	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630831
4.4.2.7.1	Choke (Electrical Test)				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Life cycles at 60 milliamps at temp		0.40%	3.9.7	Test/MIL-T-27
102	Inductance		0.40%	3.2	Test
103	Resistance		0.40%	3.2	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630840/MIL-T-27
4.4.2.8	Transformer (Visual)				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering in accordance with MIL-STD-1460 (8 places) (prior to potting)		0.40%	3.2	Visual
102	Cup height, max. (after potting)		0.40%	3.2	Gage
Minor					
201	Marking (after final electrical)		0.65%	3.10	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630840
4.4.2.8.1	Transformer (Electrical Test)				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Life cycle at 32VDC		0.40%	3.9.7	Test/MIL-T-27/ Dwg. 12630840
102	Dynamic electrical characteristic		0.40%	3.2	Test

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630860
4.4.2.9	Bridge Rectifier				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Burn-in		100%	3.9.4	Test
102	Forward voltage drop per leg		100%	3.2	Test
103	Leakage current per leg		100%	3.2	Test
Minor					
201	Marking (after final electrical)		0.65%	3.10	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.9.1	Bridge Rectifier (Environmental, Destructive Test)		1 OF 1	12630860
				NEXT HIGHER ASSEMBLY: 12630829

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Sequence of Environments Required				
Major					
101	Temperature cycling		0.40%	3.9.1	Test
102	Constant acceleration		0.40%	3.9.2	Test
103	Thermal shock		0.40%	3.9.5	Test
104	Mechanical shock		0.40%	3.9.6	Test
105	Forward voltage drop per leg		0.40%	3.2	Test
106	Leakage current per leg		0.40%	3.2	Test

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Transistor, High Power, Silicon P.N.P. (2N6193)				12630838
					NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Burn-in		100%	3.9.4	Test
102	Collector cut-off current with emitter open		100%	3.2	Test
103	Emitter-cut-off current		100%	3.2	Test
104	Collector-to-emitter sustaining voltage with base open		100%	3.2	Test
105	Collector-to-emitter saturation voltage		100%	3.2	Test
106	Base-to-emitter voltage		100%	3.2	Test
107	DC forward current transfer ratio		100%	3.2	Test
Minor					
201	Marking (after final electrical)		0.65%	3.10	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10.1	Transistor, High Power, Silicon P.N.P. (2N6193)				12630838
					NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Sequence of Environmentals Required				
Major					
101	Temperature cycling		0.40%	3.9.1	Test
102	Constant acceleration		0.40%	3.9.2	Test
103	Thermal shock		0.40%	3.9.5	Test
104	Mechanical shock		0.40%	3.9.6	Test
105	Seal		0.40%	3.9.3	Test
106	Electrical		0.40%	3.2	Test/4.4.2.10

NOTES:

AMSMC Form 1570, 1 Feb 85

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630857
4.4.2.11	Connector				NEXT HIGHER ASSEMBLY 12630854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of threaded post(2 places)		0.40%	3.2	GO/NO-GO Gage
102	Depth of threaded post (2 places)		0.40%	3.2	Gage
103	Pins damaged or missing		0.40%	3.2	Visual
104	Plating missing		0.40%	3.2	Visual
105	Foreign material on pins 1/		0.40%	3.2	Visual
106	Gasket missing or damaged 2/		0.40%	3.2	Visual
Minor					
201	Protective cap missing		0.65%	3.2	Visual
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES: 1/ Pins shall have a bright and shiny appearance.
2/ Gasket may be separated from connector.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630851
4.4.2.12	Flexible Cable				NEXT HIGHER ASSEMBLY 12630849
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Plating on pads missing		0.40%	3.2	Visual
102	Breaks in circuitry		0.40%	3.2	Visual
103	Delamination		0.40%	3.2	Visual
104	Tears or nicks in outside edge		0.40%	3.2	Visual
105	Location of holes		0.40%	3.2	Visual
106	Solder Wicking		0.40%	3.2	Visual
107	Continuity and isolation		100%	3.5.1	Test
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 12630846
4.4.2.13	Logic Circuit Board				NEXT HIGHER ASSEMBLY 12630845
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole location		0.40%	3.2	Gage
102	Hole diameter		0.40%	3.2	Gage
103	Board cracked, chipped, warped or twisted		0.40%	3.2	Visual
104	Conductive paths defects reduced by 1/3 or more of the width		0.40%	3.2	Visual
105	Conductor not bonded to board or peeling		0.40%	3.2	Visual
106	Conductor areas not plated		0.40%	3.2	Visual
107	Bridging of conductors		0.40%	3.2	Visual
108	Annular ring encirclement improper		0.40%	3.2	Visual
109	Plated through hole defects - more than three (3) plating voids per hole		0.40%	3.2	Visual 1/
110	Plated through hole defects - voids in excess of 10% of surface of hole		0.40%	3.2	Visual
111	Plated through hole defects - voids at interface of hole		0.40%	3.2	Visual
112	Plated through hole defects - separation around circumference		0.40%	3.2	Visual
113	Plating through hole thickness		0.40%	3.2	Test 2/

NOTES: 1/ A sample of three (3) holes of each diameter on each board shall be inspected.
2/ A sample of three (3) holes of each diameter on each board shall be inspected for a minimum of 1 Mil in the hole thickness.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 12630846
4.4.2.13	Logic Circuit Board				NEXT HIGHER ASSEMBLY 12630845
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Thickness		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630845
4.4.2.14	Capacitor (68 Microfarad)				NEXT HIGHER ASSEMBLY 12630845
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	Capacitance Bridge/ MIL-C-39006/22-0110
102	Dissipation factor		0.40%	3.2	Capacitance Bridge/ MIL-C-39006/22-0110
Minor					
201	Marking missing or incorrect		0.65%	3.10	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630845
4.4.2.15	Logic Circuit Board Assembly (Visual)				NEXT HIGHER ASSEMBLY 12630828
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering not in accordance with MIL-STD-1460		0.40%	3.2	Visual
102	Capacitor polarity incorrect		0.40%	3.2	Visual
103	Incorrect component, damaged or missing component		0.40%	3.2	Visual
104	Pins on wrong side of board		0.40%	3.2	Visual
105	Maximum height of soldered component leads and terminals		0.40%	3.2	Gage
Minor					
201	Height of hybrid off board		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 12630845
4.4.2.15	Logic Circuit Board Assembly (Electrical Test)				NEXT HIGHER ASSEMBLY 12630828
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Continuity and resistance		100%	3.4.1	Test
102	TM monitors (with charge command and right convertor input)		100%	3.4.2.1	Test
103	Fire TM		100%	3.4.2.2	Test
104	Arm TM		100%	3.4.2.3	Test
105	ITL TM		100%	3.4.2.4	Test
106	Convertor TM		100%	3.4.2.5	Test
107	Destruct pulse, Power Up (with Charge Command input)		100%	3.4.2.6.1	Test
108	Destruct pulse, convertor input removed Charge Command input		100%	3.4.2.6.2	Test
109	TM monitors (with Power/Destruct and left convertor input)		100%	3.4.3.1	Test

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 12630845
4.4.2.15	Logic Circuit Board Assembly (Electrical Test)				NEXT HIGHER ASSEMBLY 12630828
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110	Power TM threshold (with Power/ Destruct input)		100%	3.4.3.2	Test
111	Destruct pulse threshold (with Power/ Destruct input)		100%	3.4.3.3	Test
112	Convertor TM threshold (with Power/ Destruct input)		100%	3.4.3.4	Test
113	Five volt test, Destruct pulse		100%	3.4.4.1	Test
114	Capacitor storage		100%	3.4.2.7	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630850
4.4.2.16	Molding Compound				NEXT HIGHER ASSEMBLY 12630849
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Incorrect material		As req'd	3.2	Visual
202	Shelf life expired		As req'd	3.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630849
4.4.2.17	Connector Cable Assembly, Prior to Potting				NEXT HIGHER ASSEMBLY 12630828
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering not in accordance with MIL-STD-1460		0.40%	3.2	Visual
102	Continuity and isolation		100%	3.5.1	Test
Minor					
201	Maximum height of cable after soldering		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630849
4.4.2.18	Connector Cable Assembly, After Potting				NEXT HIGHER ASSEMBLY 12630828
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pins above potting surface		0.40%	3.2	Visual
Minor					
201	Holes or cavities in potting compound		0.65%	3.2	Visual 1/
202	Evidence of entrapped air		0.65%	3.2	Visual 1/
203	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES: 1/ Requires submission and approval of accept/reject standards.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630855
4.4.2.21	Plate, Mounting Connector				NEXT HIGHER ASSEMBLY 12630854
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of threaded holes (max) (4 places)		0.40%	3.2	Gage
102	Location of threaded holes (4 places)		0.40%	3.2	Gage
Minor					
201	Dimension of cut-out		0.65%	3.2	Gage
202	Location of cut-out		0.65%	3.2	Gage
203	Depth of recess		0.65%	3.2	Gage
204	Location of recess		0.65%	3.2	Gage
205	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630854
4.4.2.22	Connector Assembly				NEXT HIGHER ASSEMBLY 12630849
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Connector flange above flush		0.40%	3.2	Visual/Manual
102	Excess adhesive		0.40%	3.2	Visual
103	Evidence of adhesive on pins (2 sides)		0.40%	3.2	Visual
104	Orientation of connector outer shell in mounting plate assy.		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630828
4.4.2.23	Electrical Assembly (EA) Visual Prior to Conformal Coating and Adhesive				NEXT HIGHER ASSEMBLY 12630801
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering not in accordance with MIL-STD-1460		0.40%	3.2	Visual/MIL-STD-1460
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630828
4.4.2.24	Electrical Assembly (Visual After Conformal Coating)				NEXT HIGHER ASSEMBLY 12630801
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Conformal coating around mounting holes		0.40%	3.2	Visual
Minor					
201	Conformal coating not continuous		0.65%	3.2	Visual
202	Evidence of excess conformal coating		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Electrical Assembly Electrical Operation	1 OF 1	12630828
			NEXT HIGHER ASSEMBLY 12630801

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Continuity	100%	1/	3.6.1	Test
102	Resistance (Safe & Locked)	100%		3.6.2	Test
103	Connector pins to case ground isolation resistance	100%		3.6.3	Test
104	Fire circuit charging (Charge Command)	100%		3.7.1	Test
105	Fire circuit capacitor voltage storage	100%		3.7.1.1	Test
106	Fuze Fire function	100%		3.7.4.2	4.5.4/Test
107	No charging while armed	100%		3.7.3	Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630828
4.4.2.26	Electrical Assembly Electrical Operation				NEXT HIGHER ASSEMBLY 12630801
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
108	Fire circuit charging (Power/Destruct)	100%	1/	3.7.2	Test
109	Fire upon arming after loss of power	100%		3.7.4.1	4.5.2/Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12630828
4.4.2.27	Electrical Assembly Electrical Operation		NEXT HIGHER ASSEMBLY 12630801

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
110	Voltage Output (Safe and Locked)	100%	1/	3.8.2	Test
111	Voltage Output (Safe and Unlocked)	100%		3.8.2	Test
112	Voltage Output (Armed and Locked)	100%		3.8.2	Test
113	Loss of Power After Arming	100%		3.7.4.1.1	Test/4.5.1
114	Five-Volt Test (Safe and Locked)	100%		3.6.4	Test
115	Resistance (Safe and Unlocked)	100%		3.6.2	Test
116	Five-Volt Test (Safe and Unlocked)	100%		3.6.4	Test
117	Five-Volt Test (Armed and Locked)	100%		3.6.4	Test
118	Resistance (Armed and Locked)	100%		3.6.2	Test
119	Power TM Threshold	100%		3.8.1	4.5.5/Test
120	Power/Destruct Threshold	100%		3.7.4.1.3	4.5.3/Test

NOTES: 1/ Accept on 0 defects, reject on 1 or more defects.

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4.5 Methods of inspections.

4.5.1 Procedure for loss of power after arming test.

4.5.1.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. Electrical Assembly Armed and Locked.

b. 31.0 + 0.5VDC applied to Power/Destruct.

c. Left capacitor monitor @ 80.0VDC min.

d. Right capacitor monitor @ 80.0VDC min.

e. 5.0 + 1.0VDC applied to 5V test input.

4.5.1.2 Dynamic conditions.

a. Remove 31.0 + 0.5VDC on Power/Destruct.

b. Measure time (see 3.7.4.1.1) to function.

4.5.2 Procedure for fire upon arming after loss of power test.

4.5.2.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

a. Electrical Assembly Safe and Unlocked.

b. 25.2 - 0.5VDC on Power/Destruct.

c. Left capacitor monitor @ 70.0VDC min.

d. Right capacitor monitor @ 70.0VDC min.

e. 5.0 + 1.0VDC applied to 5V test input.

4.5.2.2 Dynamic conditions.

a. Remove 25.2 - 0.5VDC on Power/Destruct.

b. Arm and lock (see 3.6) electrical assembly 4.2 + 1.0 second after removal of power/destruct voltage.

c. Measure time (see 3.7.4.1.2) to function.

4.5.3 Procedure for power/destruct threshold test.

4.5.3.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

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a. Electrical Assembly Armed and Locked.

b. 23.0 ± 2.0VDC applied to Power/Destruct.

c. Left capacitor monitor @ 70VDC min.

d. Right Capacitor Monitor @ 70VDC min.

4.5.3.2 Dynamic conditions. Decrease Power/Destruct input voltage (+) J1-30 to (-) J1-15 with a voltage step no greater than 25 millivolt (mv) and record input voltage level at the time the fire circuit functions (see 3.7.4).

4.5.4 Procedure for fuze fire function.

4.5.4.1 Initial conditions. The following conditions shall exist and be verified prior to start of test:

a. Electrical Assembly Armed and Locked.

b. 23.0 - 0.5VDC applied to Power/Destruct.

c. Left capacitor monitor @ 70.0VDC min.

d. Right capacitor monitor @ 70.0VDC min.

4.5.4.2 Dynamic conditions.

a. Apply NO Fire Signal A
b. Apply NO Fire Signal B
c. Apply NO Fire Signal C
d. Apply Fuze Fire Signal

4.5.5 Procedure for power TM threshold test.

4.5.5.1 Initial conditions. The following conditions shall exist and be verified prior to start of test.

a. Electrical Assembly Armed and Locked.

b. 28.0 ± 3.0VDC applied to Power/Destruct.

c. Left capacitor monitor @ 70.0VDC min.

d. Right capacitor monitor @ 70.0VDC min.

4.5.5.2 Dynamic conditions. Decrease Power/Destruct input voltage using a voltage step no greater than 25 multivolts (mv) and record input voltage at the time the Power TM Monitor changes states (see 3.8.1).

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4.6 Tolerances.

4.6.1 Measurements (electronic). Tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

5. PACKAGING - N/A

6. NOTES

6.1. Intended use - This specification is intended to be used in conjunction with MIL-S-48687, Safety and Arming Device, GM: M143E1 Loading, Assembling and Packing For. In the event of conflict, MIL-S-48687 shall take precedence.

6.2 Ordering data - See MIL-A-48078.

6.2.1 Data requirements - Deliverable data required by this specification is cited in the following paragraphs

Paragraph	Data Requirement	Applicable DID
3.10	Test and Demonstration Reports	DI-T-1906

(Copies of data item description required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 Submission of designs for approval - See 6.2.3 to MIL-A-48078. Submit equipment designs, as required, to ATTN: AMSMC-QAR-I, Dover, New Jersey 07801-5001.

6.4 Drawings - Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is new under the cognizance of ARDEC.

6.5 Key word listing.

Safety and Arming Device
Electrical parts
Assemblies

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project: 1336-A531)

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